Evaluating Machine Learning Models for Classification of Mass Spectrometry Data

Problem Definition

Proteomics is a field in biological research, which is the study of all the proteins that can arise from gene expression in an organism. Proteomics helps uncover underlying biology in various diseases, including cancer, by studying protein expression differences between diseased and healthy states. One of the methods to generate proteomics data is by liquid chromatography-mass spectrometry (LC-MS) (**Figure 1**), where the biological sample is pre-processed down to the protein and peptide level, and the peptides are separated in a liquid chromatogram and ionized and detected as ions counted in a mass spectrometer. A recurrent issue with LC-MS proteomics spectra is the collection of chimeric spectra by the mass spectrometer. Chimeric spectra are overlapping spectra of multiple peptides originating from multiple proteins within the same scan window. Chimeric spectra confound signal and identification of proteins of interest.

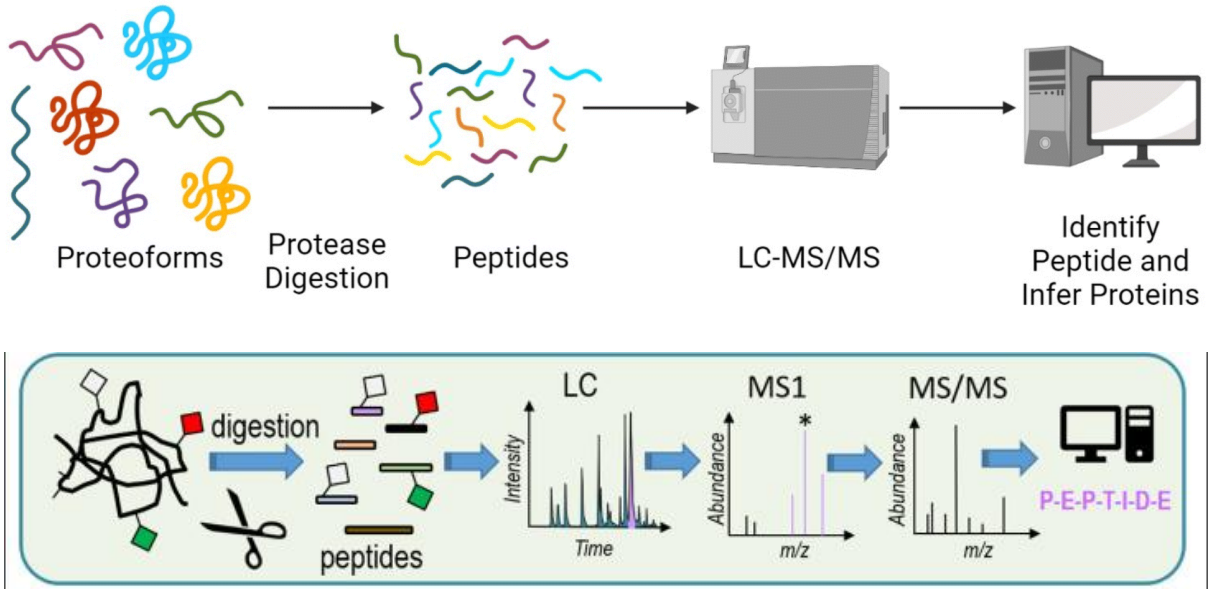

Figure 1: Proteomics samples are studied using Liquid Chromatography-Mass Spectrometry (LC-MS). The proteome of a biological sample is extracted, the proteins are chopped up into peptides and put into a liquid chromatography (LC) column. Peptides in a LC column separate and elute, with each peptide having its own retention time based on physic-chemical interactive properties of the peptide and the LC column used. The peptides are ionized before entering the mass spectrometer. During mass spectrometry (MS), the peptide ions oscillate in an electric field and are detected by the mass spectrometer as ion peaks depending on the peptide's mass-to-charge ratio. Peptide ions detected in the first round of mass spectrometry (MS1) are called precursor masses, that are selected for further fragmentation into reporter ions and give more refined spectra. Each protein produces characteristic peptide fragmentation spectral patterns. Thus, spectral data are processed using bioinformatics tools to identify the proteins present in the biological dataset.

I have a LC-MS dataset from my first-author publication that contains the *Escherichia coli* (*E. coli*) proteome spiked in with the bovine serum albumin (BSA) protein, to mimic the presence of noisy protein in the background of our *E. coli* experimental sample, and how different levels of spike-in of BSA affect *E. coli* LC-MS spectra.

Proteomics scientists are keen on developing ML/AI methods to resolve chimeric spectra, either by correcting peaks data at the MS collection level or after the LC-MS. Success in identifying chimeric spectra would help remove those messy spectral files and keep cleaner files for protein identification and data analysis at the bioinformatic level, enhancing biological discovery. Establishment of solid ML/AI approaches to resolving chimeric spectra would give greater insights on improving LC-MS proteomics experiment protocol and technical features of mass spectrometers for cost-efficient use of resources in R&D. A useful outcome from one such ML/AI model would be the ability to identify and delete chimeric spectral files without losing clean spectral files, allowing for accurate bioinformatics analysis of LC-MS proteomic datasets.

The goal of this project was to develop a suitable ML model that can differentiate complex MS signals of peptides as sample complexity, i.e., more BSA spike-in, increases.

Approach and Tool Selection

Data pre-processing

I converted spectral files into tabular data frames containing information of both meta-data and actual peaks data. The files are large with > 12 million rows listing each peak detected in each scan window recorded by the mass spectrometer. I collapsed these large original data frames into data frames containing features to train the ML model. Working with these large files was mainly guided by ChatGPT prompting to load and manipulate on Python without Python crashing.

I commenced this project with an initial proxy label classification of chimeric and non-chimeric spectra. As I have got datasets containing 50ng *E. coli* proteome with no (0ug) BSA, 5ug BSA and 50ug BSA, I created proxy labels of ‘non-chimeric’ to 0ug BSA and ‘chimeric’ to 5ug and 50ug of BSA. Working at the Linear Discrimination Analysis (LDA) classification and Random Forest (RF) levels did not yield good models at the peaks-level data.

Hence, I re-started my project by working at the meta-data level, and tweaked the goal of my project. Instead of proxy label creation of ‘chimeric’ and ‘non-chimeric’, I decided to train my datasets to identify the labels of 0, 5 and 50ug of BSA experimental condition labels, which would imply the model’s ability to match spectral features to biological sample noise, which would be a indicative of spectral chimericity, as it is known that more noisy biological samples lead more chimeric spectra.

As seen in the Jupyter notebook, I adjusted data type settings from the default ones that Python works with in order to save RAM.

ML model selection

Feature Engineering: After loading the data sets, I created a features table after some advice from ChatGPT on which LC-MS features were most relevant for ML modelling, especially since I have got tabular data frames.

I started out the ML modelling with unsupervised clustering, doing a Principal Component Analysis (PCA) to check whether the selected features correlated the BSA condition at all. Here is the result:

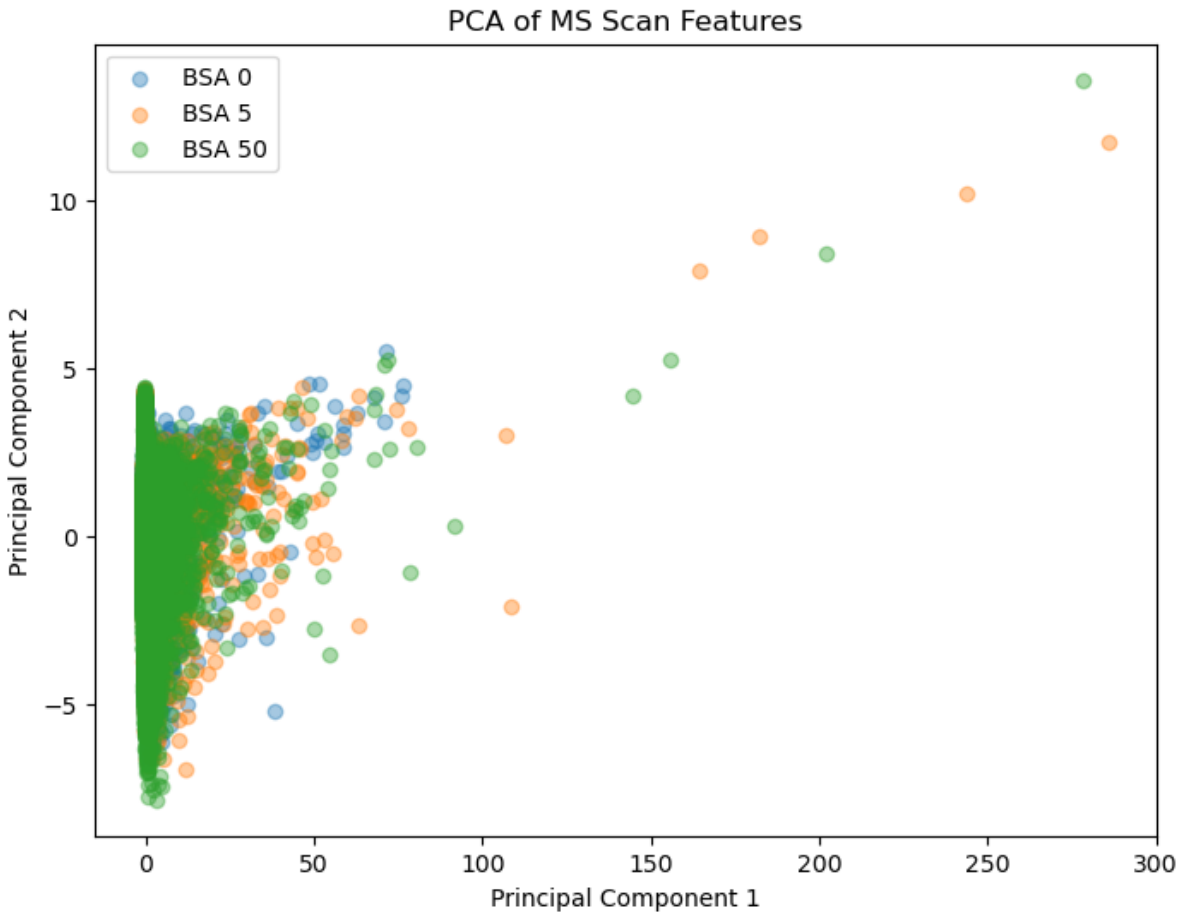

Figure 2: **Principal Component Analysis (PCA) of engineered features with BSA condition (0ug, 5ug and 50ug of BSA).** The PCA plot shows a long diagonal structure with features forming a continuous gradient, and some features from 5ug and 50ug of BSA labels are stretching outwards to the top-right of the plot. There is no real formation of clusters and there is heavy overlap between the engineered features.

As seen in figure 2, there is no distinguishable variance between features regarding BSA condition. This is not surprising, as the concept of chimeric spectra has been shown to be difficult if entirely tested upon sample contamination. Other sources of variance giving rise to chimeric spectra include retention time patterns, precursor mass ranges (the mass of the peptide at MS1 level, see **Fig. 1**) and scan complexity.

I moved on to do supervised ML through Linear Discriminant Analysis. Compared to PCA, LDA might mathematically account for both in-class and between-class variance, which might allow for better separation in the presence of labels 0, 5 and 50ug BSA. The LDA classification resulted in about 40% accuracy, with a relatively consistent precision between experimental labels, but showed decreasing trend in recall with increasing amounts of BSA – 50ug BSA had the lowest recall score. This shows that the variance in features is not strongly linear with relationship to the experimental condition. Increase in contaminated spike-in to the biological sample seems to affect the LDA classification more poorly.

Given that my data is in tabular form and that LDA showed that features don't correlate in a particularly linear fashion, I chose to do a Random Forest (RF) model on my data. Random Forests are commonly used in biological research as they are known to capture feature interactions thus help uncover complex biological datasets including LC-MS data.

The initial RF model gave 57% in accuracy, crossing 50% in precision and recall for the data, except recall for 50ug BSA (47%). Overall, the f-1 score also crossed 50%, indicating usefulness of using a RF model to explain variance in features of complex LC-MS proteomics datasets. The RF model shows that feature interaction with experimental condition label is better explained to be a non-linear relationship than a linear one, in comparison with results of the LDA.

To finalise which ML model to use, I compared the RF model with Logistic Regression (LR) and Gradient Boosting (GB), which work with tabular data that have features with non-linear variance. LR produced a model score of about 40% accuracy; in particular, recall score for 0ug BSA experimental condition stands at 0%, deeming the LR model as inappropriate for ML of the datasets as LDA. GB produced a lower accuracy score (51%) than RF, thus RF was selected as the appropriate ML method for my project.

RF model finalization

I performed hyperparameter tuning scoring to refine the RF model. Given how large my datasets are, I tweaked the parameter distances from the default for hyperparameter tuning, allowing for 18 model trainings instead of 30 with 3-fold cross-validation than the usual 5. The results of the hyperparameter tuning suggest that the RF model would run successfully at 100 trees, 1 leaf, 2 splits and a depth of 20 branches without overfitting the data, producing a 57% accuracy score exactly as the initial RF model.

Cross-validation (CV) of the chosen RF model at 3-fold cross-validation rather than 5 and using RandomisedSearchCV was done to prevent RAM exhaustion, as advised by ChatGPT. CV produced a mean accuracy score of 45%, suggesting that the model may be overfitting, or that performance varies across different data splits. When we repeated cross-validation on the training dataset only rather than the full data, the mean accuracy remains unchanged, suggesting that most of the variance in the data was captured in the training dataset.

Reflection

ChatGPT was used to design the ML pipeline. Originally, I was told to do LDA classification on my dataset during grad school. However, as this didn't work and I also don't have true labels of chimeric and non-chimeric spectra in my dataset, ChatGPT guided my approach through trial-and-error in choosing the right ML model.

The RF model performed best than other ML models tested, with an accuracy score better than expected, given that LC-MS biological data is noisy with complex spectral patterns. Having achieved 57% accuracy in this RF model suggests meaningful contribution of LC-MS features to spectral complexity. Exploring feature importance in the RF model (**Figure 3**) shows that retention time contributed the most (~ 16%) to the model. It is well-known in the field of LC-MS proteomics that retention time can be manipulated to explore the possibilities of noisy chimeric LC-MS spectral files.

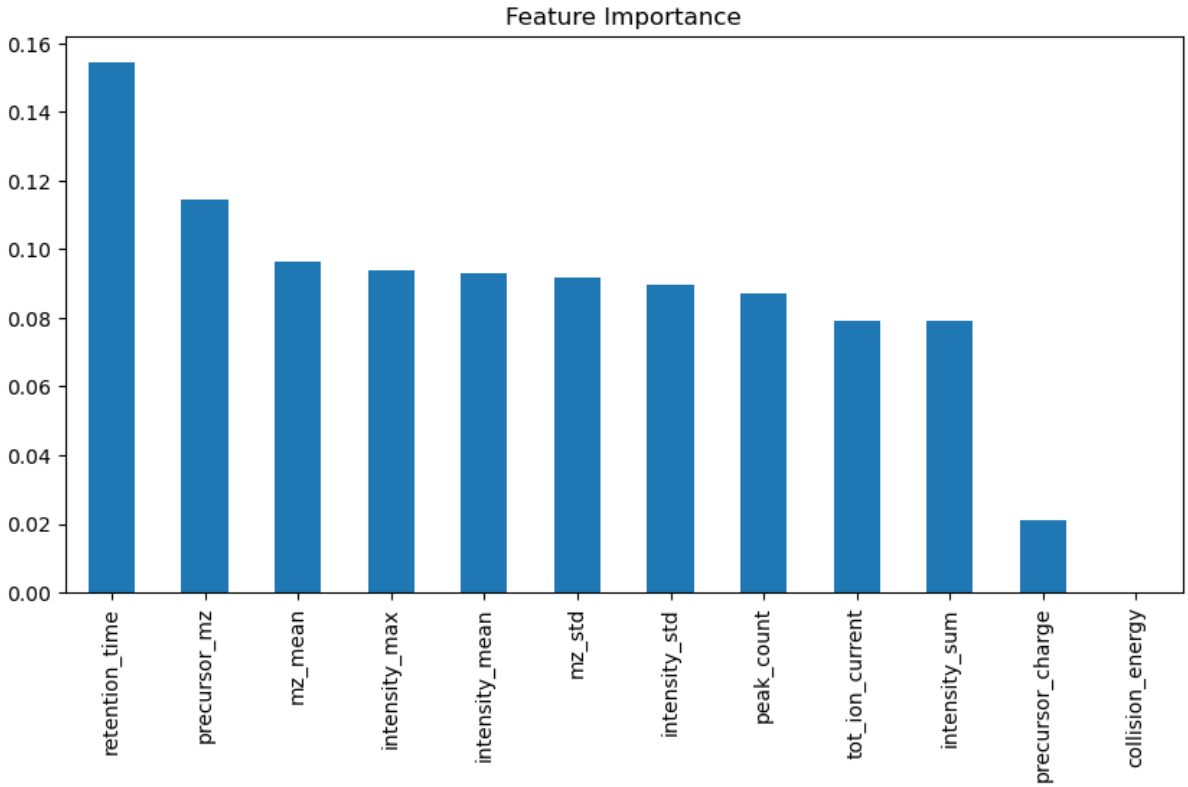

Figure 3: Importance of features to the Random Forest model to identify experimental condition. Retention time was the most important feature that contributed to variance in the dataset, followed by precursor mass (>10% importance), mean of m/z masses, peptide signal intensity and peak counts (all < 10%). Collision energy does not help distinguish classes.

The main challenge of machine learning LC-MS-based proteomics datasets is the variance of complexity of the biological sample and the technical variability between MS scan methods, making features difficult to train into a ML model. As seen with this RF model, at higher concentrations of BSA, i.e. contamination spike-in, the dataset becomes noisier thus features become more difficult to train.

A major challenge with this project was dealing with the very large sizes (>4GB) of my LC-MS datasets. I sized down the data types on Python to prevent RAM usage on my laptop and prevent Python from crashing. A better approach to dealing with large files like these would be to work with them in cloud-based environments, where Python is available as a virtual module in clouds that can be loaded to write a job-scheduling script to work with such large files. Working on a cloud cluster might help overcome current limitations from limited hyperparameter search and cross-validation that was used to build the RF model.

This RF model could be improved by adding more meaningful features at the meta-data level through statistical manipulation of primary LC-MS features, including retention time. Further literature review/proteomics researcher expert input would be taken into consideration for stronger feature engineering of such large-scale, complex LC-MS biological datasets.

Development of an appropriate ML to predict the relationship of LC-MS features with experimental conditions would help identify and remove chimeric spectra at the LC-MS level, so fully clean spectral files can be obtained at the MS level and thus can be used to process at the protein identification and subsequent bioinformatics stages, which would make the

data analytics process of large-scale biological datasets, including proteomics data, easier and thus accelerate biological discovery. ML approaches would also provide insights to mass spectrometer companies to account for chimeric spectra chances when scanning complex biological samples.